We established KidneyGenAfrica to facilitate multi-centre research in Africa, enhance large-scale genomic study and deliver research and training excellence in genomics of kidney function in Africa

KidneyGen Africa

KidneyGenAfrica Core Team

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Co-Director, PI
MRC Uganda & QMUL
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Bioinformatics
Uganda, UK

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Michele Ramsay
Wits, South Africa

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MRC Uganda

Mia Crampin
MEIRU, Malawi

Oyekanmi Nash
CGRI/NABDA, Nigeria

Adebowale Adeyemo
NIH, USA

https://www.kidneygenafrica.org/

Funding: UKRI/MRC (2024-2029)

Be a Partner

- **What we offer**
  - Potential to host an African ECR
  - Access to training opportunities for your students and postdoctoral researchers
  - Opportunities to be included in new publications
  - Opportunities to be included partnership grant applications, enabling partners to co-develop and lead competitive funding proposals
  - Speaking or facilitation opportunities at workshops, symposia, and stakeholder meetings
  - Visibility on KidneyGenAfrica platforms (website, newsletters, reports, events)
  - Opportunities to lead or co-lead work packages within multi-country projects

- **What we expect**
  - Genetic dataset linked with kidney function
  - Contribution to KidneyGenAfrica training and capacity building

Current collaborators – Potential Host for Research Exchange

Research exchanges (potential hosts)

Collaborator	Institution Name
Dorothea Nitsch & Laurie Tomlinson	LSHTM
Andrew P Morris	University of Manchester
Claudia Langeberg & Segun Fatumo	Queen Mary University of London
Anna Köttgen	Institute of Genetic Epidemiology University Medical Center Freiburg
Eleftheria Zeggini	Institute of Translational Genomics in Helmholtz Zentrum München
Cristian Pattaro	Eurac Research, Bolzano
Nora Franceschini	University of North Carolina at Chapel Hill
Charles Rotimi & Adebowale Adeyemo	NIH
Cassianne Robinson-Cohen	Department of Medicine at Vanderbilt University Medical Center

Queen Mary University of London

MRC/UVRI and LSHTM Uganda Research Unit

Medical Research Council

Uganda Virus Research Institute

London School of Hygiene & Tropical Medicine

Polygenic Risk Score in Africa: Application, Challenges and Future Direction

Segun Fatumo PhD, FHEA, FAAS
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Queen Mary University of London, United Kingdom

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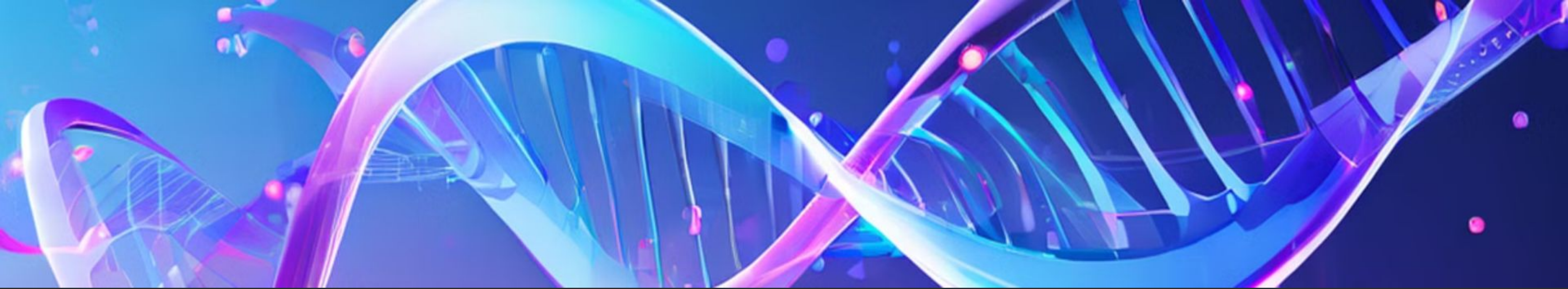
Entebbe, Uganda

wellcome connecting science

KidneyGen Africa

Outline

- ✔ Polygenic Risk Score – What is it and how is it computed ?
- ✔ Transferability of trans-ancestry eGFR GRS into Uganda
- ✔ Genetic risk scores of eGFR in KidneyGenAAfrica
- ✔ Type 2 Diabetes: Genetic Prediction of Type 2 Diabetes in Continental Africa
- ✔ Lipid Traits: Transferability of genetic risk scores of lipid traits in Africa
- ✔ PRS: The Challenges and Potential Future for Precision Medicine in Africa



What is a Polygenic Risk Score?

Polygenic risk scores (PRS) are a powerful tool in the field of personalized genomics, providing insights into an individual's predisposition to complex diseases by analyzing their genetic makeup.

1 Genetic Predisposition

PRS quantifies an individual's genetic risk for a specific disease or trait, taking into account the combined effect of many genetic variants.

2 Complex Diseases

PRS are particularly useful for understanding the genetic basis of complex diseases, which are influenced by multiple genes and environmental factors.

3 Personalized Medicine

PRS can be used to identify individuals at high risk for certain diseases, enabling earlier intervention and personalized preventive strategies.

How is a Polygenic Risk Score Calculated?

1

Genome-Wide Association Studies (No PRS without GWAS)

Genome-Wide Association Studies (GWAS) identify genetic variants associated with specific traits or diseases.

2

Weighting and Summing

PRS are calculated by summing the effects of multiple genetic variants, weighted by their associated risk estimates from GWAS.

3

Statistical Modeling

Advanced statistical techniques, such as machine learning, are used to refine the PRS model and improve its predictive accuracy.

- **Output from GWAS analysis is a Summary Statistics**
- Most important columns are:
  - Chr: Chromosome
  - SNP: SNP Id
  - BP: Basepair
  - EA: Effect Allele
  - NEA: Non Effect Allele or Reference Allele
  - EAF: Effect Allele Frequency
  - Effect: Odd ratio or Beta
  - SE: Standard Er
  - P: Pvalue

risk allele: in the context of a disease, this is the allele that confers a risk of developing the disease. Most of the time, risk allele = minor allele, as most people will not carry the risk allele. However, in some case, the risk allele can in fact be the major allele.

Chr	SNP	bp	A1	A2	Freq	b	se	p	K
1	rs2286139	761732	C	T	0.1379	-0.0104056	0.00732416	0.155397	0.5
1	rs12562034	768448	A	G	0.10475	-0.00627592	0.00827054	0.447955	0.5
1	rs4970383	838555	A	C	0.247975	0.00946201	0.00587444	0.107243	0.5
1	rs1806509	853954	C	A	0.3912	0.0152744	0.00523012	0.00349507	0.5
1	rs13302982	861808	A	G	0.018025	-0.0180122	0.0189517	0.341895	0.5
1	rs28576697	870645	C	T	0.29355	0.0116486	0.00556379	0.0362916	0.5
1	rs2340582	882803	A	G	0.05465	0.0119371	0.0111055	0.282426	0.5
1	rs3748594	886384	A	G	0.025975	-0.01244	0.0158797	0.433401	0.5
1	rs28504611	908414	T	C	0.022225	0.00388796	0.0171623	0.820781	0.5
1	rs9777939	929190	A	G	0.03345	-0.00446279	0.0141522	0.752502	0.5
1	rs1891910	932457	A	G	0.2295	-0.00647527	0.00605864	0.285175	0.5
1	rs35940137	940203	A	G	0.050575	-0.10689	0.0115935	2.97533e-20	0.5
1	rs6657048	957640	T	C	0.011825	0.0892934	0.0233322	0.000129691	0.5
1	rs9803031	987200	C	T	0.083875	-0.00434284	0.0091306	0.634334	0.5

Computing a PRS

- ❑ Single variant association analysis has been the primary method in GWAS but requires very large sample sizes to detect more than a handful of SNPs for many complex traits

- ❑ In contrast, PRS analysis does not aim to identify individual SNPs but instead aggregates genetic risk across the genome in a single individual polygenic score for a trait of interest.

- ❑ In this approach, a large discovery sample is required to reliably determine how much each SNP is expected to contribute to the polygenic score (“weights”) of a specific trait.

- ❑ Subsequently, in an independent **target** sample, which can be more modest in size, polygenic scores can be calculated based on genetic DNA profiles and these weights

Working example of three single nucleotide polymorphisms (SNPs) aggregated into a single individual polygenic risk score (PRS).

Discovery GWAS

	Weight*	Risk Allele
SNP1	0.2	A
SNP2	-0.3	C
SNP3	0.1	G

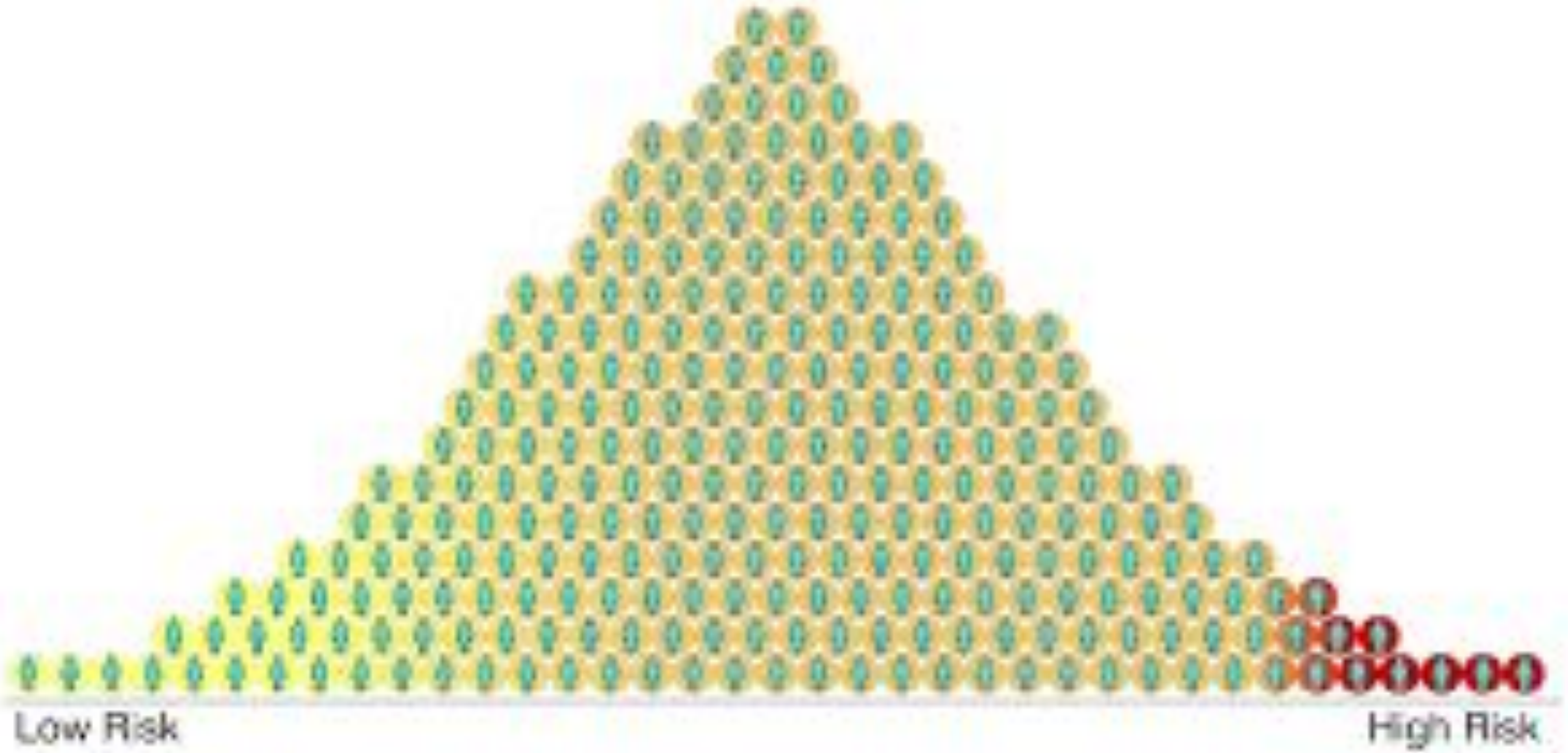
Individual	Alleles SNP1	Alleles SNP2	Alleles SNP3
1	AT	AA	CG
2	AA	CA	GG
3	TT	AC	CG
4	TT	AA	GG
5	TA	CA	GC
6	AT	CA	CG
7	AA	AA	GG
8	AA	CC	CG
9	TA	CC	GC
10	AT	AA	CG

PRS:

Individual	SNP 1	SNP 2	SNP 3	**PRS**
1	0.2+0.0	0.0+0.0	0.0+0.1	**0.3**
2	0.2+0.2	-0.3+0.0	0.1+0.1	**0.3**
3	0.0+0.0	0.0-0.3	0.0+0.1	**-0.2**
4	0.0+0.0	0.0+0.0	0.1+0.1	**0.2**
5	0.0+0.2	-0.3+0.0	0.1+0.0	**0.0**
6	0.2+0.0	-0.3+0.0	0.0+0.1	**0.0**
7	0.2+0.2	0.0+0.0	+0.1+0.1	**0.6**
8	0.2+0.2	-0.3-0.3	0.0+0.1	**-0.1**
9	0.0+0.2	-0.3-0.3	0.1+0.0	**-0.3**
10	0.2+0.0	0.0+0.0	0.0+0.1	**0.3**

- ❑ To conduct PRS analysis, trait-specific weights (beta's for continuous traits and the log of the odds ratios for binary traits) are obtained from a discovery GWAS.
- ❑ In the target sample, a PRS is calculated for each individual based on the weighted sum of the number of risk alleles that he or she carries multiplied by the trait-specific weights.
- ❑ For many complex traits, SNP effect sizes are publicly available (e.g., https://www.med.unc.edu/pgc/downloads)
- ❑ Although in principle all common SNPs could be used in a PRS analysis, it is customary to first clump (see **clumping**) the GWAS results before computing risk scores
- ❑ *p* value thresholds are typically used to remove SNPs that show little or no statistical evidence for association

$$PRS_i = \sum_{j}^{M} \hat{\beta}_j \times dosage_{ij}$$



Conducting polygenic risk prediction analyses

- Once PRS have been calculated for all subjects in the target sample, the scores can be used in a (logistic) regression analysis to predict any trait that is expected to show genetic overlap with the trait of interest.

- The prediction accuracy can be expressed with the (pseudo-)R^2 measure of the regression analysis

- It is important to include at least a few PCA or MDS components as covariates in the regression analysis to control for population stratification.

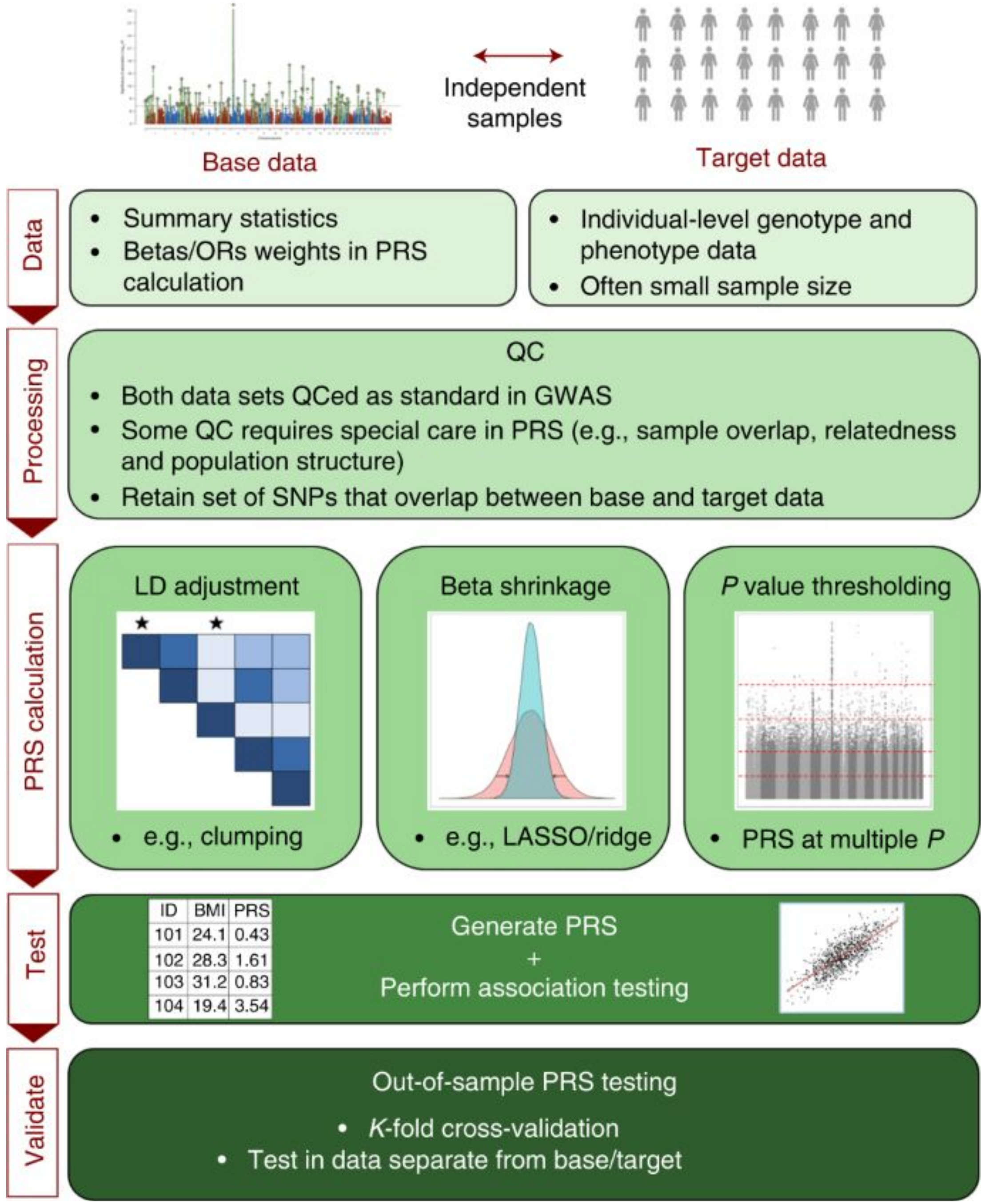
- To estimate how much variation is explained by the PRS, the R^2 of a model that includes only the covariates (e.g., MDS components) and the R^2 of a model that includes covariates + PRS will be compared.

- The increase in R^2 due to the PRS indicates the increase in prediction accuracy explained by genetic risk factors.

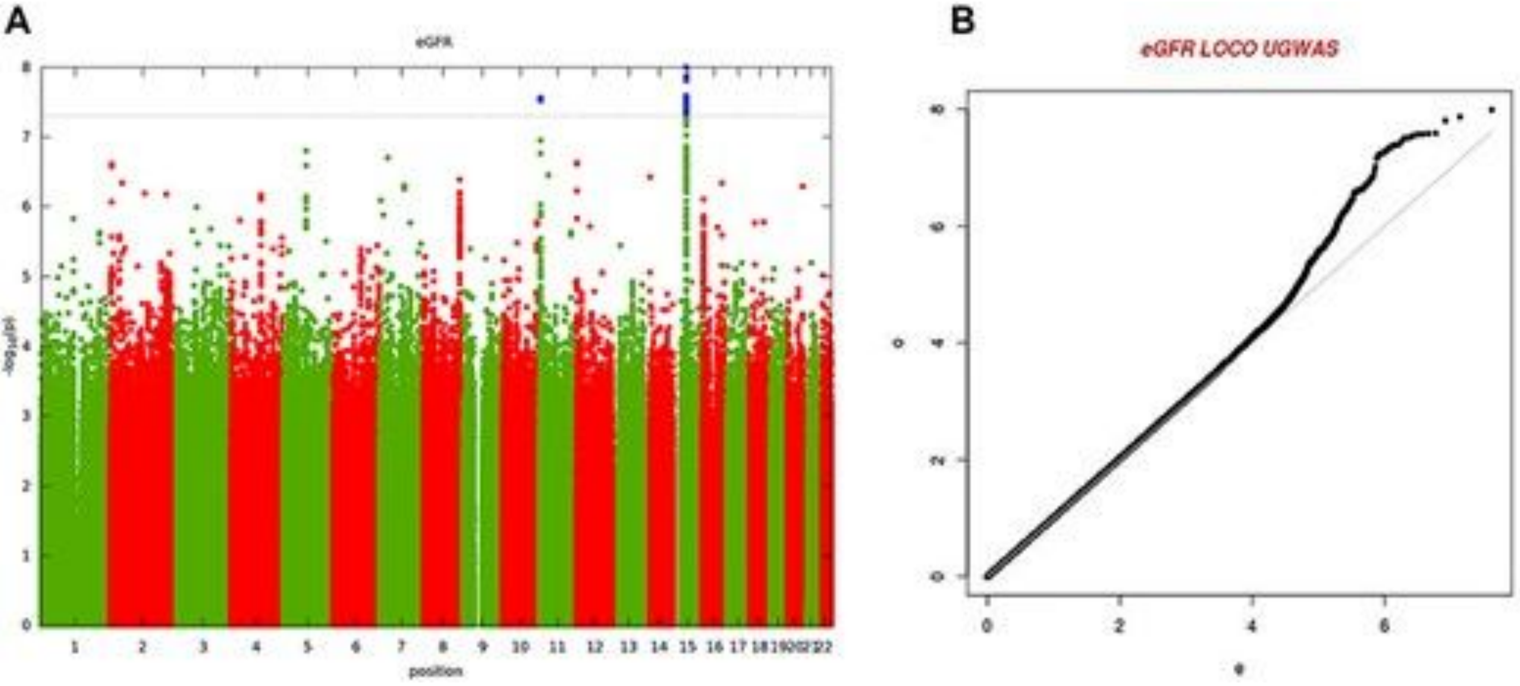
PRS analysis USING SOFTWARE PRSice

- A convenient program to perform PRS analysis is PRSice. It takes care of clumping, *p* value thresholds, MDS components, and plots attractive graphs.

- There are many other programs for the application of PRS eg,
  - LDPred, JAMPred, SBLUP, LDpred2-Inf, bigsnpr, LDpred-funct, LDpred2, Lassosum, BridgePRS, PRS-CS, PRS-CSx-auto, SBayesR, MegaPRS and even PLINK (--score))



First GWAS of kidney function (eGFR) identified and validated two eGFR loci



Fatumo S et al, 2021 Human Mol Genetic

3288 participants for UGR and replication in 8224 African Americans from the Women's Health Initiative (WHI).

Identified and validated two eGFR loci. ***GATM and HBB*** in the Uganda with 3288 participants

The lead SNP at the ***HBB*** **locus accounted for 88% of the posterior probability of causality** after fine-mapping.

Genetic risk score

- No GWAS of Kidney function in continental Africa, so we identified the largest trans-ancestry GWAS of kidney function

n=312,468
127 associated loci signals
Morris AP et al 2019

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Article | Open access | Published: 03 January 2019

Trans-ethnic kidney function association study reveals putative causal genes and effects on kidney-specific disease aetiologies

Andrew P. Morris, Thu H. Le, Haojia Wu, Artur Akbarov, Peter J. van der Most, Gibran Hemani, George Davey Smith, Anubha Mahajan, Kyle J. Gaulton, Girish N. Nadkarni, Adan Valladares-Salgado, Niels Wacher-Rodarte, Josyf C. Mychaleckyj, Nicole D. Dueker, Xiuqing Guo, Yang Hai, Jeffrey Haessler, Yoichiro Kamatani, Adrienne M. Stilp, Gu Zhu, James P. Cook, Johan Ärnlöv, Susan H. Blanton, Martin H. de Borst, ... Nora Franceschini + Show authors

Nature Communications **10**, Article number: 29 (2019) | Cite this article

n=1,046,070 264
associated loci
Wuttke et al 2019

nature genetics

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Article | Published: 31 May 2019

A catalog of genetic loci associated with kidney function from analyses of a million individuals

Matthias Wuttke, Yong Li, Man Li, Karsten B. Sieber, Mary F. Feitosa, Mathias Gorski, Adrienne Tin, Lihua Wang, Audrey Y. Chu, Anselm Hoppmann, Holger Kirsten, Ayush Giri, Jin-Fang Chai, Gardar Sveinbjornsson, Bamidele O. Tayo, Teresa Nutile, Christian Fuchsberger, Jonathan Marten, Massimiliano Cocca, Sahar Ghasemi, Yizhe Xu, Katrin Horn, Damia Noce, Peter J. van der Most, Lifelines Cohort Study, V. A. Million Veteran Program, ... Cristian Pattaro + Show authors

Nature Genetics **51**, 957–972 (2019) | Cite this article

Transferability of trans-ancestry eGFR GRS into Uganda

- Lead SNPs from trans-ancestry a meta-analysis of eGFR (Morris et al 2019) was used to evaluate the predictive power of an **unweighted GRS into unrelated individuals in the Uganda population (n=3288)**.

- Removed 817 first-degree relatives from the Uganda GPC cohort derived from PIHAT values > 0.5, we calculated principal components using –pca in PLINK

- PLINK - The predictive power of the GRS was evaluated by assessing the change in R^2 (variance explained) when it was added to the linear model of eGFR adjusted for age, sex and principal components.

Trans-ancestry GRS was not significantly associated with eGFR in the Uganda population ($P = 0.076$) and accounted for only 0.04% of the trait variance after accounting for age, sex and principal components to adjust for population structure but showed the correct direction of effect

Regression coefficients for the association of the GRS with eGFR in the Ugandan population

Variable	Model without GRS			Full model		
	Beta (SE)	P	R^2	Beta (SE)	P	R^2
Age	−0.92 (0.01)	2.0×10^{-16}	0.5492	0.93 (0.015)	2.0×10^{-16}	0.5496
Sex	−2.53 (0.56)	7.9×10^{-6}		−2.55 (0.53)	6.8×10^{-6}	
PC1	22.62 (21.53)	0.294		21.77 (21.53)	0.312	
PC2	−40.46 (22.01)	0.066		−41.78 (22.01)	0.058	
PC3	32.57 (20.17)	0.106		33.18 (20.16)	0.099	
PC4	24.34 (23.62)	0.303		24.62 (23.61)	0.297	
PC5	−2.95 (21.05)	0.889		−3.40 (21.04)	0.872	
GRS				−0.17 (0.09)	0.076	

- We were **unable to undertake a weighted GRS** because different transformations of the trait were performed in the trans-ancestry meta-analysis and in the GPC GWAS

Association of weighted GRS derived from Wuttke et al with eGFR in Uganda

Regression coefficients for the association of weighted GRS derived from Wuttke *et al.* (10) with eGFR in the Ugandan population

Variable	Model without GRS			Full model		
	Beta (SE)	***P***	***R^2***	**Beta (SE)**	***P***	***R^2***
Age	−0.92 (0.01)	2.0×10^{-16}	0.5513	−0.92 (0.01)	2.0×10^{-16}	0.5512
Sex	−2.45 (0.56)	1.49×10^{-5}		−2.44 (0.56)	1.53×10^{-5}	
PC1	42.86 (14.81)	0.294		42.88 (14.81)	0.003	
PC2	−11.08 (14.82)	0.454		−10.60 (14.84)	0.475	
PC3	42.23 (14.80)	0.004		42.42 (14.81)	0.004	
PC4	−9.59 (14.81)	0.517		−10.17 (14..83)	0.492	
PC5	29.97 (14.82)	0.043		30.02 (14.82)	0.043	
GRS				−12.38 (19.43)	0.524	

- Assessed the predictive power of a weighted GRS into unrelated individuals in the Uganda population leveraging Wuttke et 2019 in over 1million trans-ancestry meta-analysis.
- This GRS also showed the correct direction of effect though not significantly associated with eGFR in the Uganda population (*P* = 0.524) and accounted for only 0.01% of the trait variance after accounting for age, sex and principal components to adjust for population structure

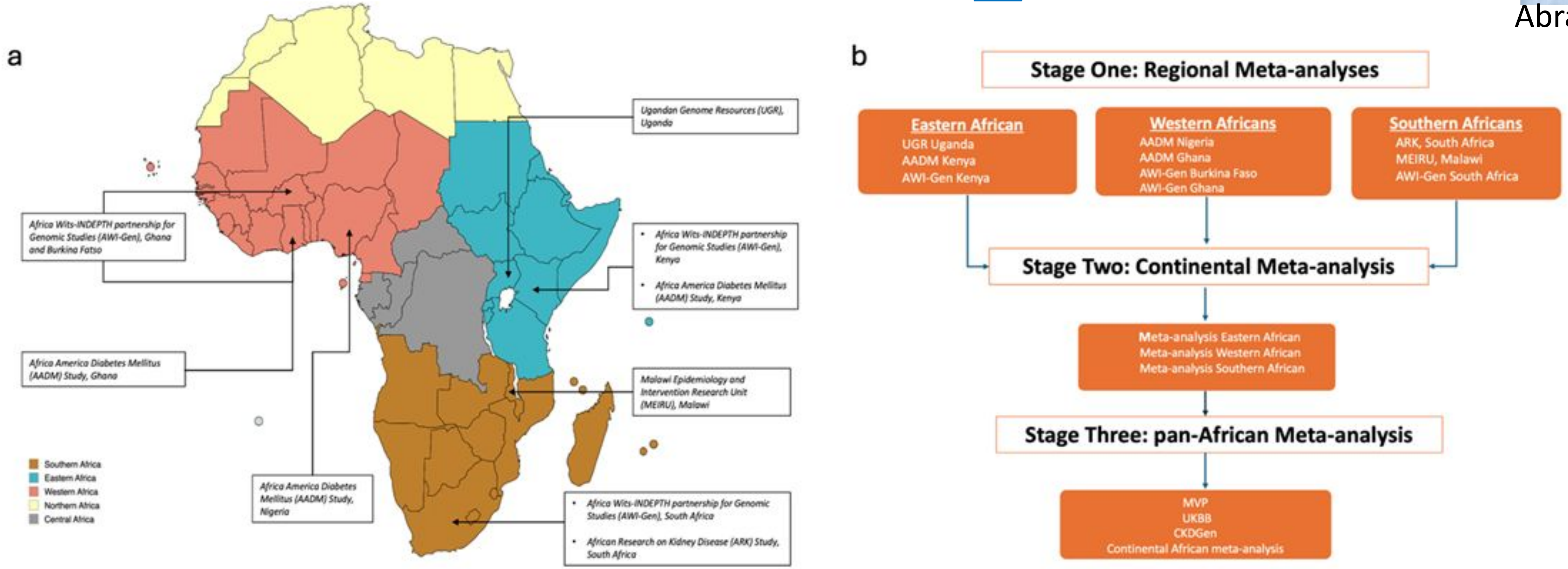
Why was GRS derived from previously reported trans-ancestry lead SNPs for eGFR not significantly predictive in Uganda ?

- This is mostly likely due to a lack of power in GPC because of small sample size.

- The lack of transferability could also be because of the way eGFR is calculated in continental Africa. Studies have shown that there is potential error measurement of serum creatinine in continental Africa that might lead to inaccurate estimates of kidney disease at individual and population level

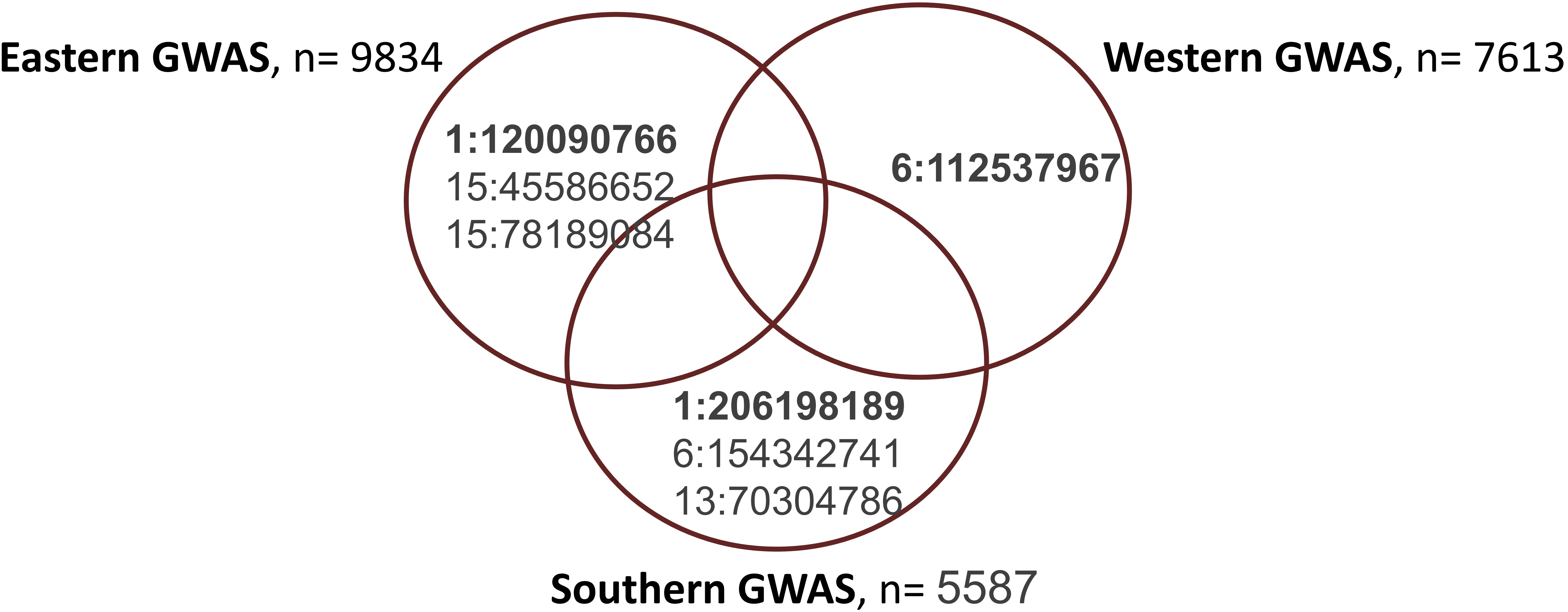
KidneyGenAfrica 1: Overall GWAS Design

Abram Kamiza



Map of Africa showing geographical regions of studies included in KidneyGenAfrican genome-wide association study meta-analysis. ARK south Africa(n=1060), AWI-gen, south Africa(n=4527), MEIRU(n=6380), UGR(n=6407), AWI-Gen, Kenya(n=1704), AADM, Kenya (n=2069), AWI-Gen, Ghana and Burkina Faso (n=3475), AADM, Ghana (n=1060), AADM, Nigeria(n=2069)

Distinct genetic factors may influence eGFR in continental Africa: No loci overlap across the different geographical regions in Africa



Stage One: Regional GWAS Meta-analysis

<u>Associated loci were defined following stringent criteria</u>.

- It reaches genome-wide significance ($P < 5 \times 10^{-8}$) in the regional meta-analysis and nominal significance ($P<0.05$) in at least two of the contributing cohorts within that region.

- It shows a consistent direction of effect across all contributing cohorts in the region.

Table 1 Independent loci associated with eGFR in individuals of African ancestry in the regional meta-analysis

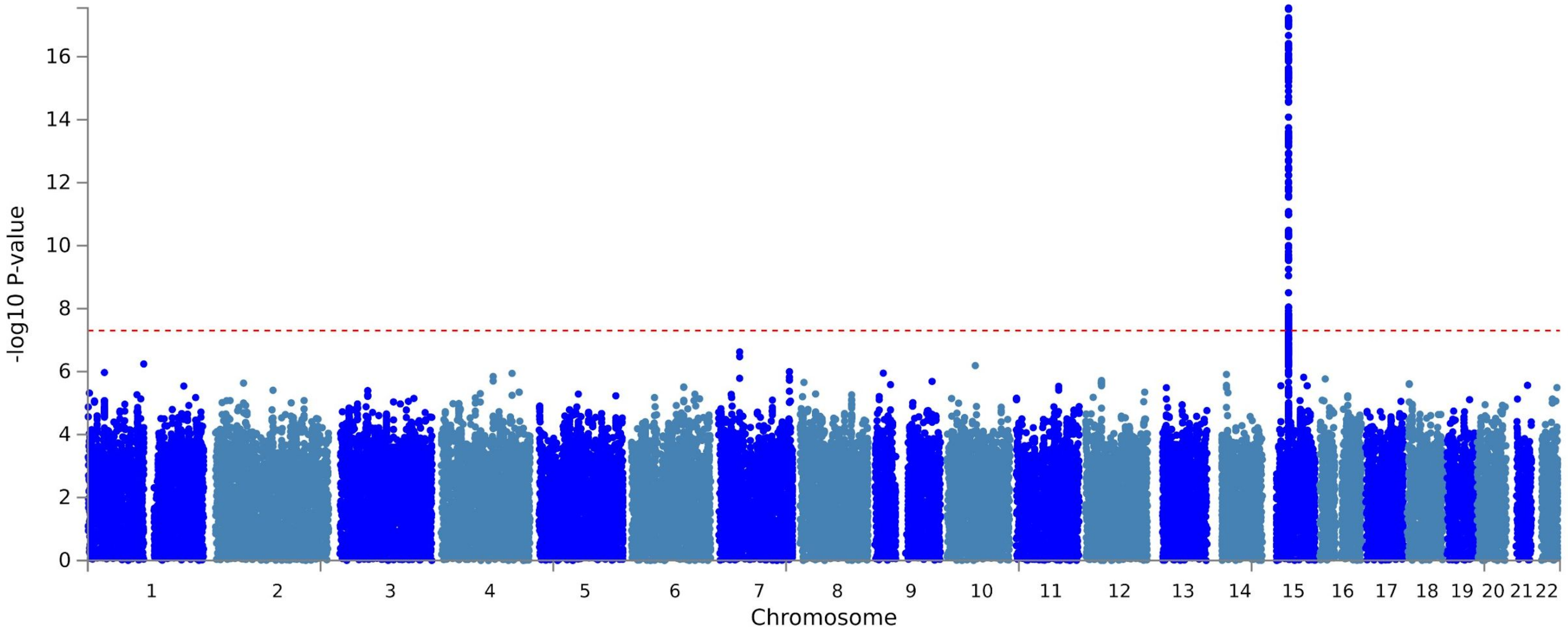
SNP	Genes	CHR	BP	EA	NEA	BETA	SE	P-value	Region	EAF-AFR†	EAF-EUR‡	EAF-EAS‡
rs6670659	*HSD3BP3*	1	120090766	C	G	0.11	0.019	1.24E-08	Eastern	0.769	0.00	0.00
rs73788952*	*OPRM1*	6	154342741	G	A	0.132	0.022	5.04E-09	Southern	0.092	0.00	0.00
rs1706775	*SLC28A2*	15	45586652	T	C	0.133	0.015	3.85E-17	Eastern	0.572	0.00	0.00
rs4243062*	*LOC645752*	15	78189084	T	C	-0.101	0.018	3.19E-08	Eastern	0.299	0.06	0.13

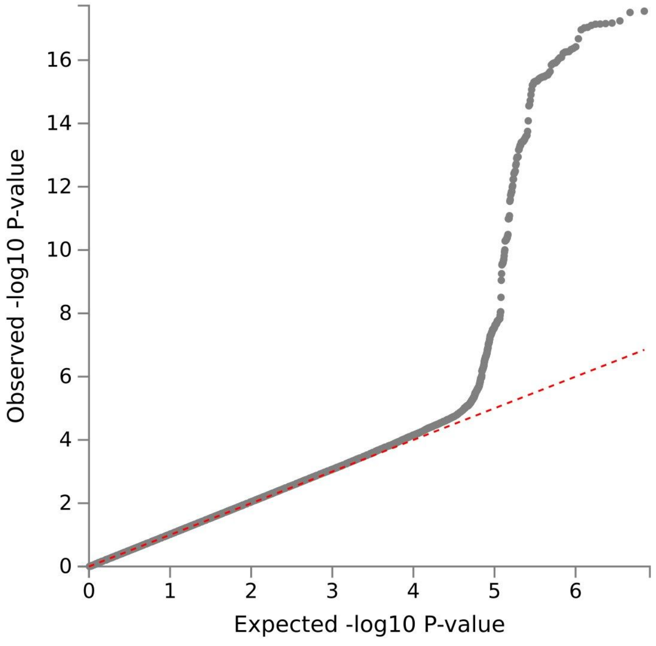
SNP; single nucleotide polymorphism, CHR chromosome, BP, base pair position, EA; effect allele, NEA, non-effect allele, EAF; effect allele frequency, SE; standard error, * novel loci, AFR; African, EUR; European, EAS; East Asian ancestry, †effect allele frequency from this meta-analysis, ‡effect allele frequency from 1000 genome project data.

1

Stage Two: Continental Meta-analysis GWAS

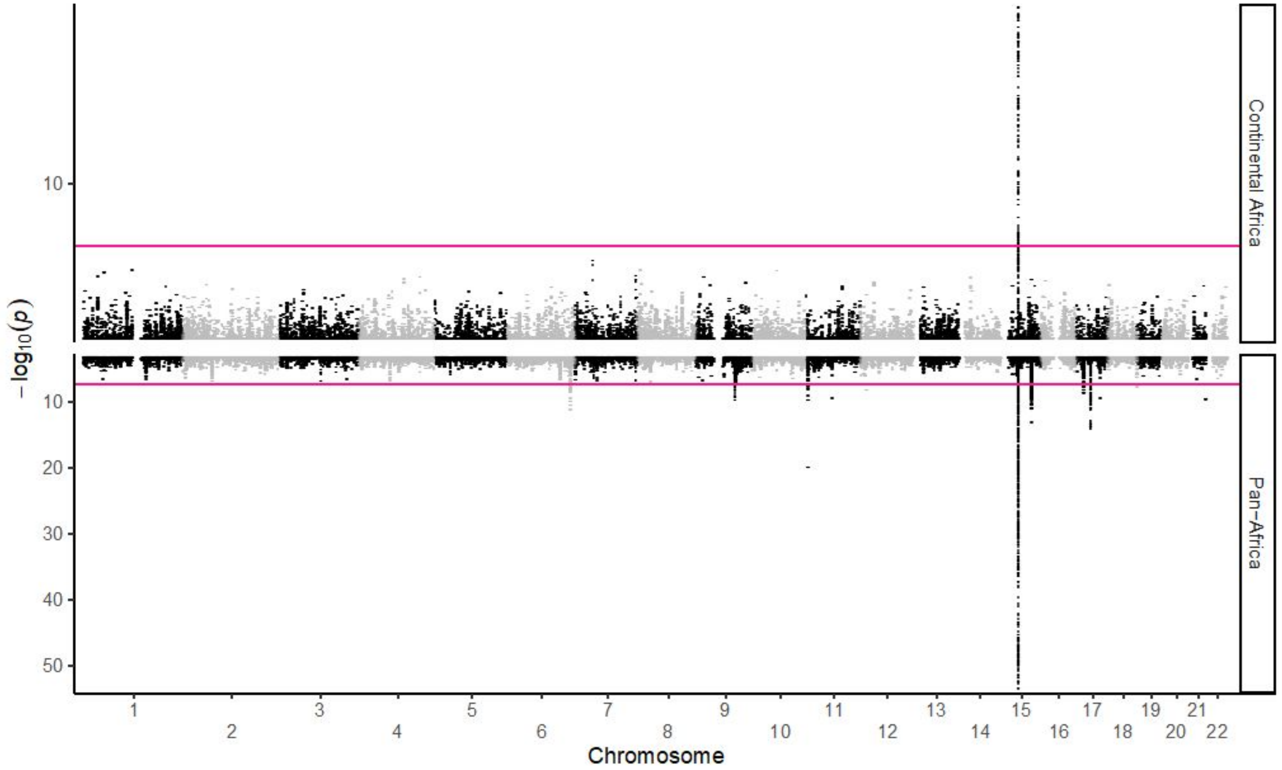
Very strong *GATM* signal at chr15 but also **observed attenuation of regional eGFR signal**

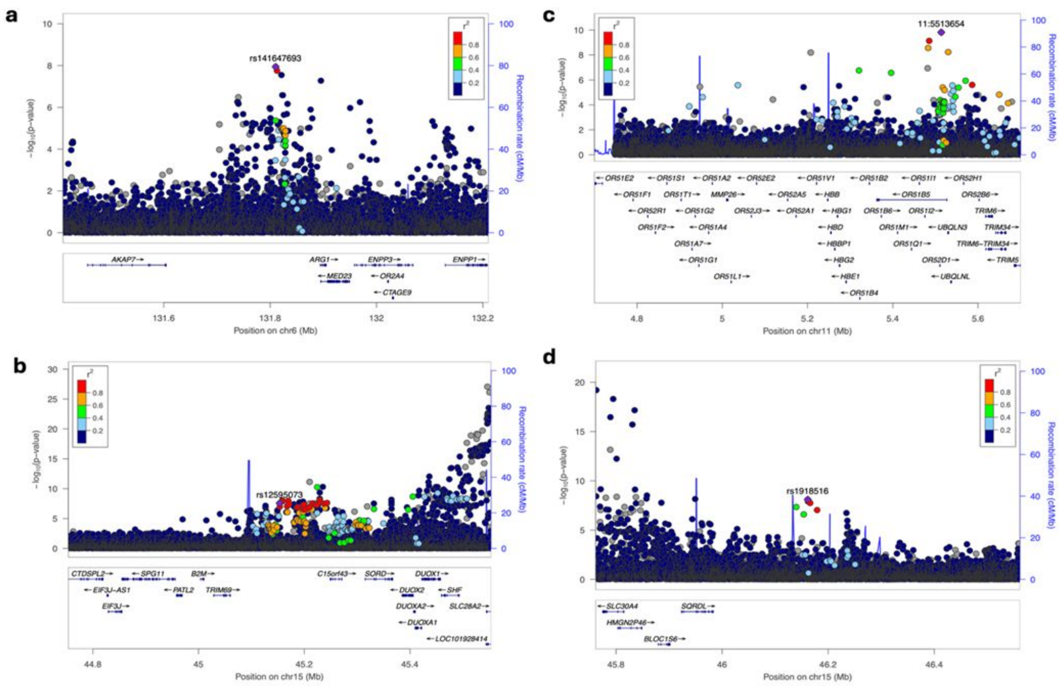




Stage Three: Pan-Africa GWAS Meta-Analysis

Pan-Africa eGFR GWAS identified 20 independent loci including **four novel association** (*ARG1, OR52H1, TRIM69,* and *SQRDL*).





Genetic Prediction of eGFR kidney function in Africa

Discovery (Base) Datasets

- *eGFR **Southern African** GWAS (n= 5587)*
- *eGFR **Western African** GWAS (n= 7613)*
- *eGFR **Eastern African** GWAS (n= 9834)*
- *eGFR **Continental African** GWAS (n= 23,034)*
- *eGFR **Diaspora African** GWAS (n= 80,027)*
- *eGFR **Pan-African** GWAS (n= 103,061)*

Target Dataset

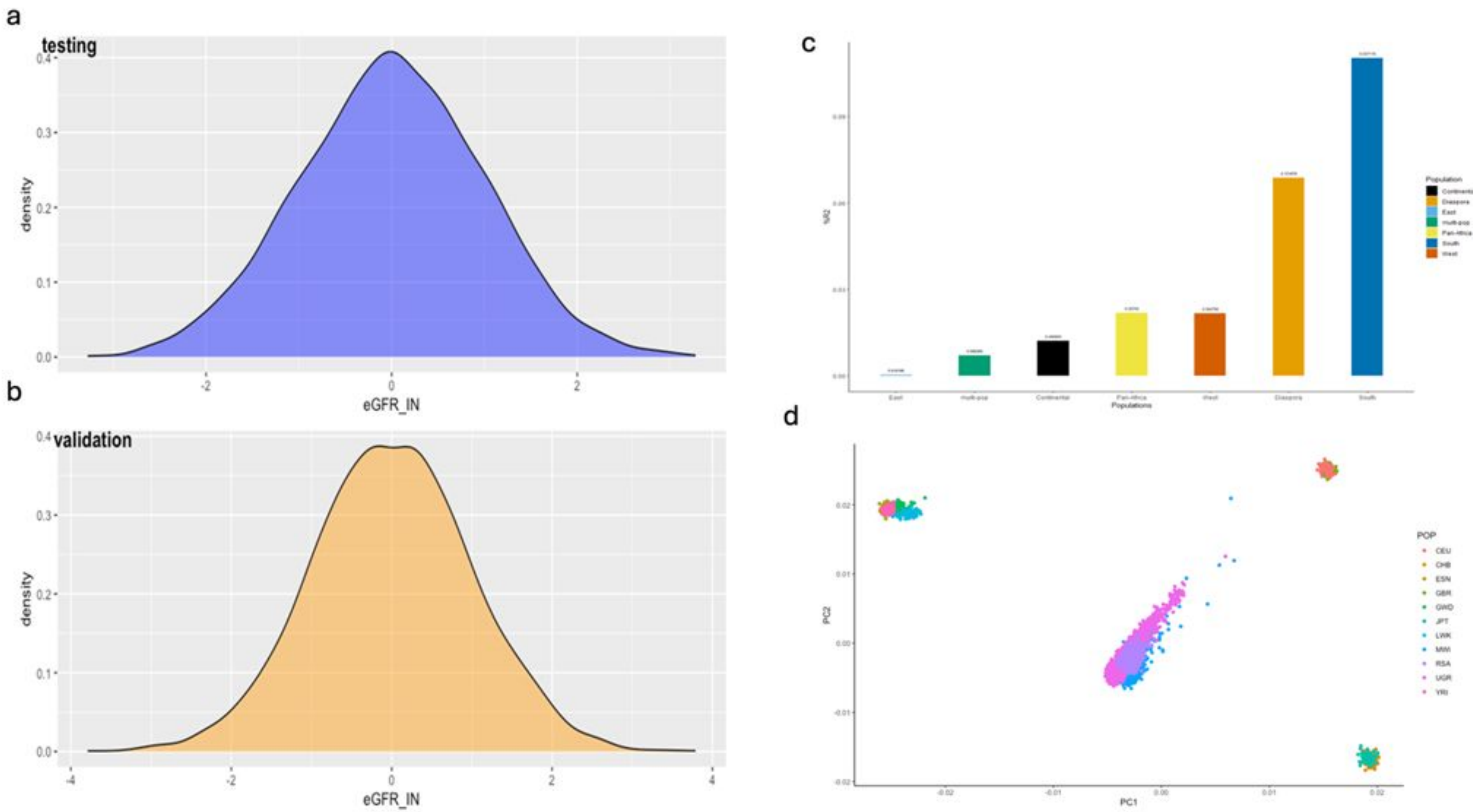
Malawi
(*n*=3099)

Validation Dataset

Malawi
(*n*=3099)

C+T Approach PRS using PRSice software

Predictivity and Transferability of Genetic Risk Score of kidney function may be driven by genetic distance in African Populations



Fine-mapping Analysis identified four loci with a single SNP accounting for >0.99 posterior probability of causality

- We established **causal missense variants -** rs334 in *HBB* and rs624307 in *SLC25A45* which are common in African populations but absent/very rare in other populations.

Jenn Asimit

Fine-Mapping: Genomic regions with variants with high posterior probability of causality

Region	Nearest Gene(s)	Variants with PP > 0.90	MAF	Most severe consequence	PP	CS99 size
7:786192	*UNCX, MICALL2*	rs13230509	0.30	regulatory region variant	0.97	2
11:4748232	***HBB***	**rs334**	**0.06**	**missense variant**	**1.00**	**1**
11:64644075	***SLC25A45***	**rs624307**	**0.09**	**missense variant**	**1.00**	**1**
15:75669004	*UBE2Q2*	rs8034430	0.11	Intron	0.97	3
17:58956589	***BCAS3***	**rs9895661**	**0.49**	**noncoding transcript exon variant**	**1.00**	**1**
18:76660235	***NFATC1***	**rs56376587**	**0.19**	**Intron**	**1.00**	**1**

PP; posterior probability of causality, MAF; minor allele frequency, CS99; 99% credible set size

Role of *APOL1* haplotypes in low eGFR in continental Africa

- In African Americans, the *APOL1* G1 and G2 variants are genetic variations strongly associated with an increased risk of developing chronic kidney disease (CKD) with approximately **13% of African American population carries these variants**.

- In our study, we observe an overall frequency of APOL1 high-risk haplotypes in continental Africa to be ~5%, ***significantly lower than the approximately 13% observed in African Americans***. In different African regions, we observe substantial regional variation in APOL1 allele frequencies with the highest prevalence of high-risk genotypes in **South African (8.6%) cohorts**, and the lowest in **East Africa (1.3%)**

- Notably, we found a **limited association between *APOL1* variants and low eGFR in continental Africa**, contrasting with the strong *APOL1* association with low eGFR seen in African Americans, highlighting distinct genetic risk profiles for kidney disease between African populations and African Americans.

Oyesola Ojewunmi

KidneyGenAfrica multi-cohort Genome-wide association study and polygenic prediction of kidney function in 110,000 Africans

Abram B. Kamiza[1,2,3,4,5*], Tinashe Chikowore[6,7,8*], Guanjie Chen[9], Oyesola Ojewunmi[1,2], Tafadzwa Machipisa[10,11], Feng Zhou[1,12], Richard Mayanja[1,13], , Sounkou Toure[14], Opeyemi Soremekun[15,1], Christopher Kintu[1,2,16], Mariam Nakabuye[1,16,17], Mine Koprulu[2], Allan Kalungi[1,2], Robert Kalyesubula[16], Babatunde Salako[18], Oyekanmi Nashiru[19], Manuel Corpas[20,21], Cassianne Robinson-Cohen[22], Nora Franceschini[23], Cristian Pattaro[24], Anna Köttgen[25], Dorothea Nitsch[4], Claudia Langenberg[2,26], Catherine Tcheandjieu[27,28], Moffat Njirenda[1,4], Andrew P Morris[29], Jennifer Asimit[12], Eleftheria Zeggini[15,30], Charles Rotimi[9], Michele Ramsay[5], Adeyemo Adebowale[9], June Fabian[31,32], Amelia C. Crampin[3,33,34,35], Jean-Tristan Brandenburg[5,36], Segun Fatumo[2,1,4,#]

"Accepted in Principle" Nature Communication

KidneyGenAfrica Research 2

KidneyGenAfrica – Research II

Identify and characterise genetic variants that influence kidney function in African-ancestry populations, and determine how these are different from those of other global populations

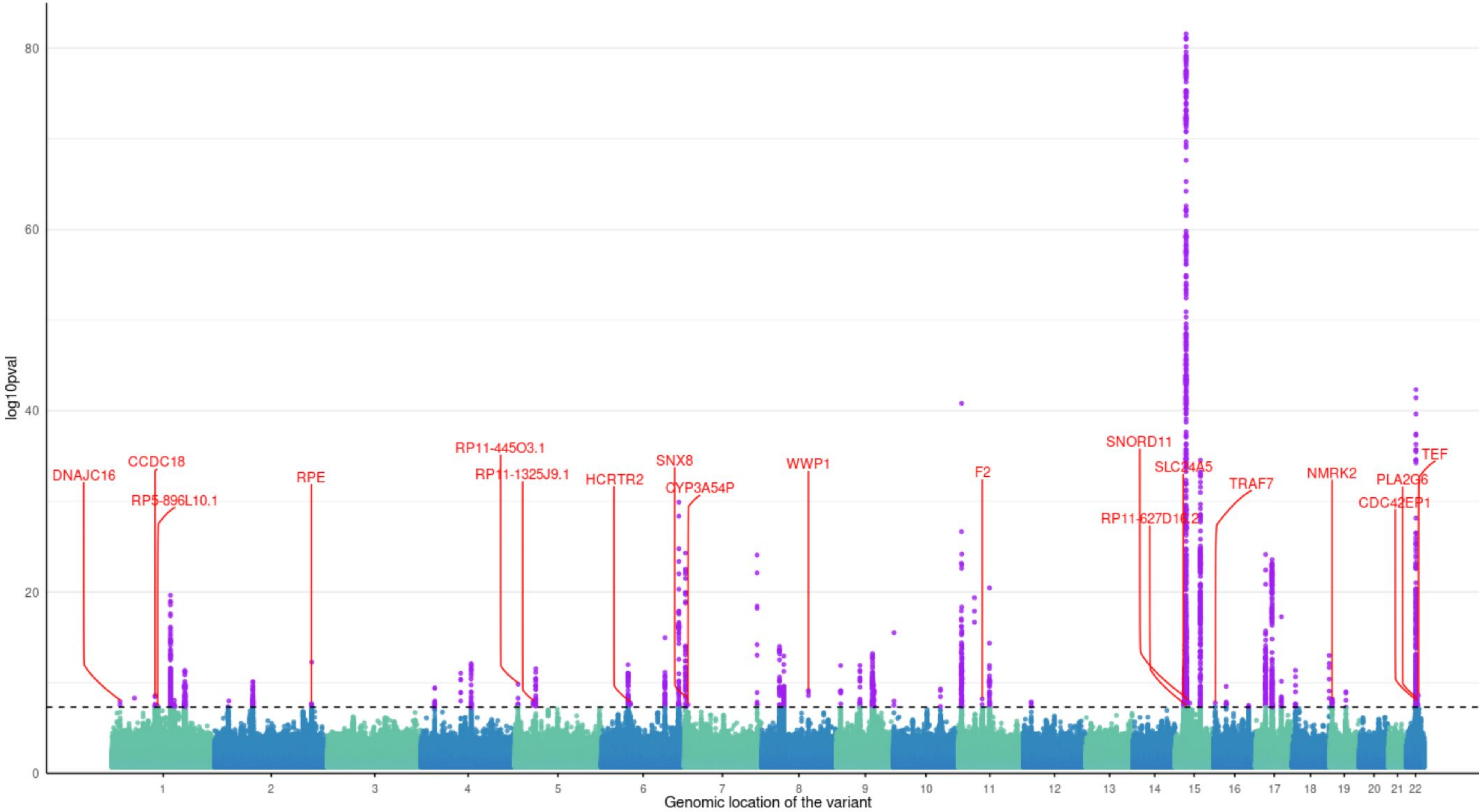
A. Continental Africa

	Study	Study Acronym	Country	Number of participants	Genotyping platform
1	Uganda Genome Resource	UGR	Uganda	6,407	Illumina HumanOmni 2.5M SNP array
2	Malawi Epidemiology and Intervention Research Unit	MEIRU	Malawi	6,380	Infinium H3Africa 2.5M SNP array
3	The Africa Wits-INDEPTH partnership for Genomic Studies	AWI-GEN	Burkina Faso, Ghana, Kenya, and South Africa	9,706	Infinium H3Africa 2.5M SNP array
4	The African Research on Kidney Disease	ARK	South Africa	1,060	Infinium H3Africa 2.5M SNP array
5	Africa America Diabetes Mellitus Study	AADM	Ghana and Nigeria	5,198	Affymetrix Axiom PANAFR and Illumina MEGA array
	Sub-total			**28,751**	

B. Genetically Inferred African Ancestry in the UK and US

	Study	Study Acronym	Ethnicity	Number of participants	Genotyping platform
1	Million Veteran Programme	MVP	Genetically inferred African ancestry	110,850	Affymetrix Axiom Biobank array
2	UK Biobank and Epidemiology Network-Kidney consortium	COGENT-Kidney	Genetically inferred African ancestry	44,025	Multiple arrays PMID: 38190104
3	All of Us	All of Us	Genetically inferred African ancestry	37,124	Illumina Global Diversity Array (GDA)
4	Vanderbilt University Medical Center's BioVU	BioVU	Genetically inferred African ancestry	To be communicated	To be communicated
	Sub-total			**191,999**	
	Total			**220,750**	

We identified previously unidentified 19 genomic loci associated with eGFRcreatinine in African individuals



Oyesola Ojewunmi

Manhattan plot representing the results from the meta-analysis for eGFR in Kidneygen Africa. The black dashed line represents the genome-wide significance threshold at p<5E-8. Any genetic variant that passes the significance threshold has been coloured blue. The signals not identified in previous studies have been annotated by their closest gene in red.

Scientists need to WALK the TALK

UK Biobank (n=~500,000)

~38,000 individuals with non-European ancestry

92.7% of publications excluded non-EUR samples

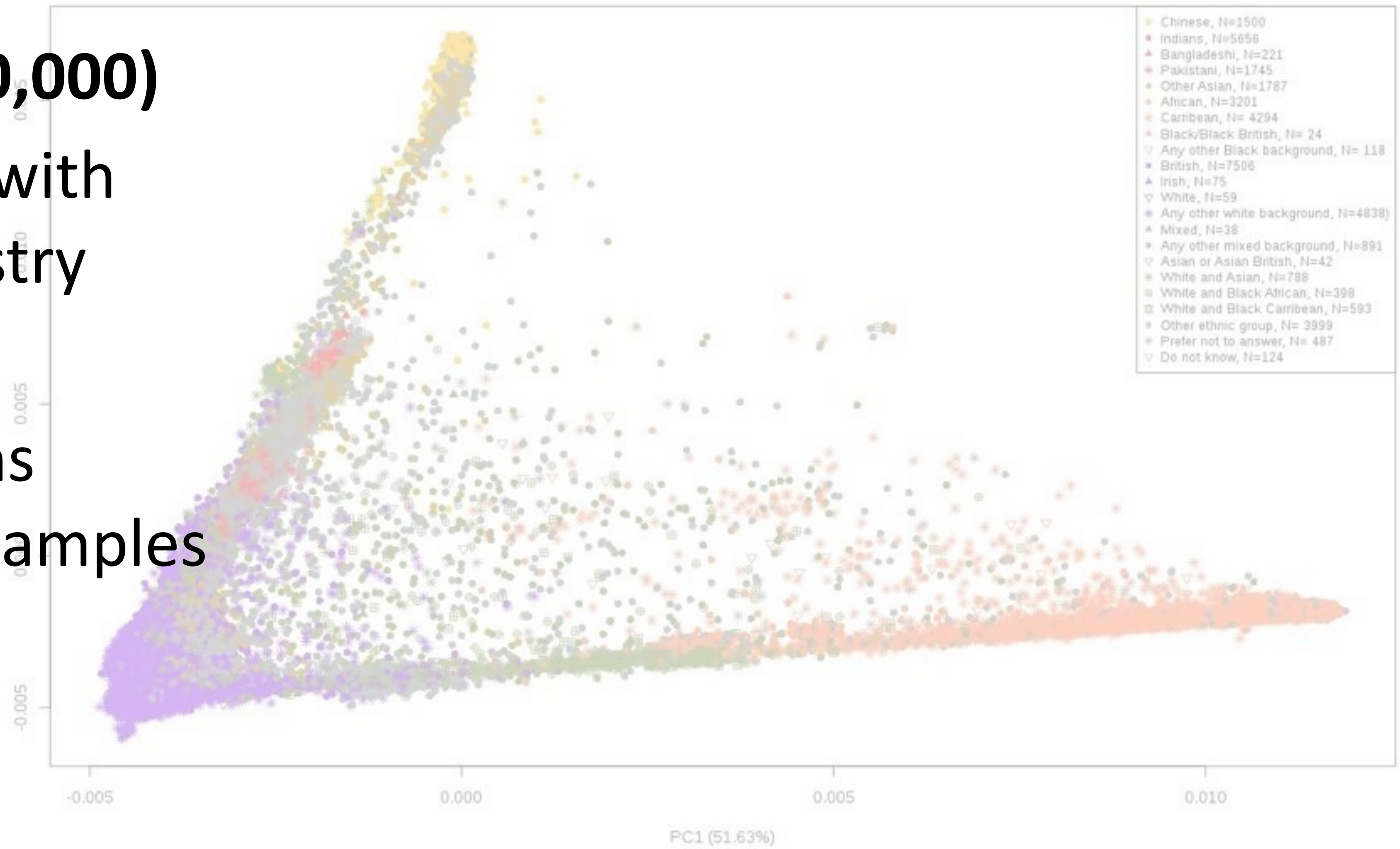


Diagram credit: **Karoline Kuchenbaecker**

- Genetic Prediction of **Type 2 Diabetes** in Continental Africa

Diabetes Care. 2022 Mar; 45(3): 717–723.
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PMCID: PMC8918234
PMID: 35015074

Polygenic Prediction of Type 2 Diabetes in Africa

Tinashe Chikowore,[✉] Kenneth Ekoru, Marijana Vujkovi, Dipender Gill, Fraser Pirie, Elizabeth Young, Manjinder S. Sandhu, Mark McCarthy, Charles Rotimi, Adebowale Adeyemo, Ayesha Motala, and Segun Fatumo[✉]

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Study design

- **Discovery (Base) Datasets**

  - *T2D **Multi-ancestry** meta-analysis GWAS (n= 1,407,282)*
  - *T2D **African American** GWAS (n = 56,092)*
  - *T2D **European** GWAS (n= 1,114,458)*

C+T Approach PRS using PRSice software

We set out to evaluate best PRS to improve polygenic prediction in continental Africans.

Target Dataset

South Africa – Zulu
(*n=2,583*)

Validation Dataset

AADM
(*n=4,309*)
Kenya – East Africa
Nigeria – West Africa
Ghana – West Africa

Table 1 Comparisons of the predictive ability of ethnically derived PRS on type 2 diabetes in continental Africans

	Multi-ethnic	African American	European
Discovery Dataset (Multi-ancestry meta-analysis)			
Cases	228,499	24,646	148,726
Controls	1,178,783	31,446	965,732
PRS Development			
Target Data Set (SA Zulu)			
Cases	1,602	1,602	1,602
Controls	981	981	981
PRS parameters			
P-value threshold	3×10^{-4}	5×10^{-8}	0.0608
Number of SNPs	41,815	65	405,572
Nagelkerke R2 %	0.69	1.11	0.69
P-value	4.62×10^{-6}	3.90×10^{-9}	5.09×10^{-6}
*OR(95%CI)	1.29 (1.16-1.43)	1.58 (1.36-1.84)	1.01 (1.00-1.01)
*P-value	3.52×10^{-6}	4.80×10^{-9}	9.54×10^{-6}
Validation of PRS			
Validation data set (AADM)			
Cases	2148	2148	2148
Controls	2161	2161	2161
PRS parameters			
P-value threshold	3×10^{-4}	5×10^{-8}	0.0608
Number of SNPs	41,553	65	1,408,065
Nagelkerke R2 %	2.62	2.92	0.13
P-value	1.06×10^{-21}	9.38×10^{-24}	2.99×10^{-2}
*OR(95%CI)	1.04 (1.03-1.05)	1.57 (1.47-1.67)	1.004 (1.03-1.05)
*P-value	1.41×10^{-21}	5.91×10^{-23}	3.16×10^{-2}

*models adjusted for ancestry indicated by 5 principal components, age, sex and BMI; OR = odds ratio; CI = confidence interval.

- The best predictive PRS from the data on Europeans, the multi-Ancestry individuals, and African Ancestry was significant

- The highest variance as indicated by Nagelkerke R^2 values of
    - **Europeans: 0.69%**
    - **multi-Ancestry: 0.69%**
    - **African Ancestry: 1.11%**

- The best PRSs were validated in the AADM study and were noted to be all significant in a similar trend

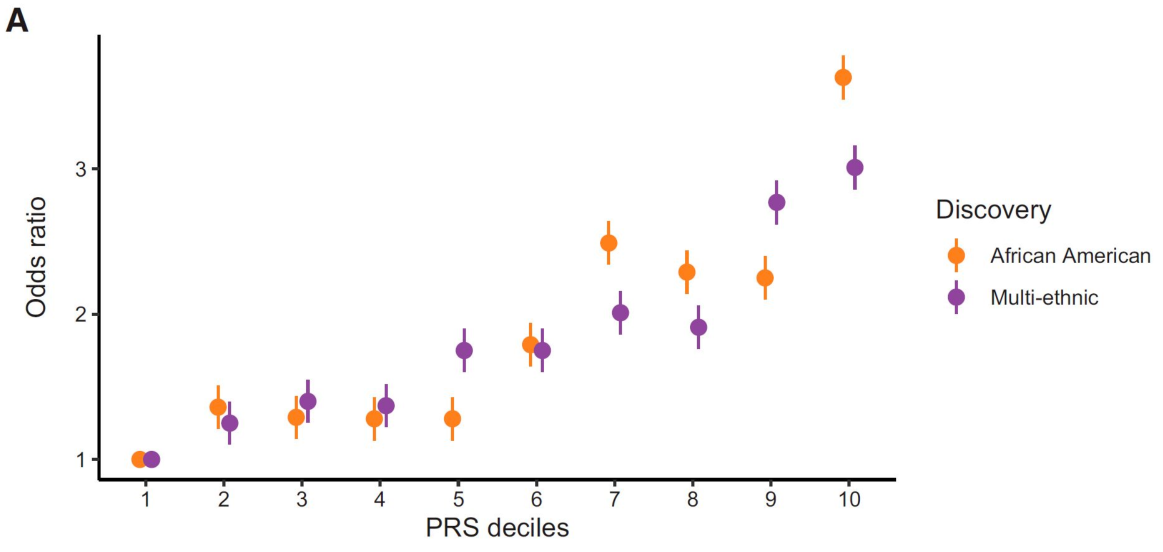
- The African Ancestry PRS had the highest predictability, indicated by a Nagelkerke R^2 of 2.92% (9.38×10^{-24}) in the combined analysis of the countries

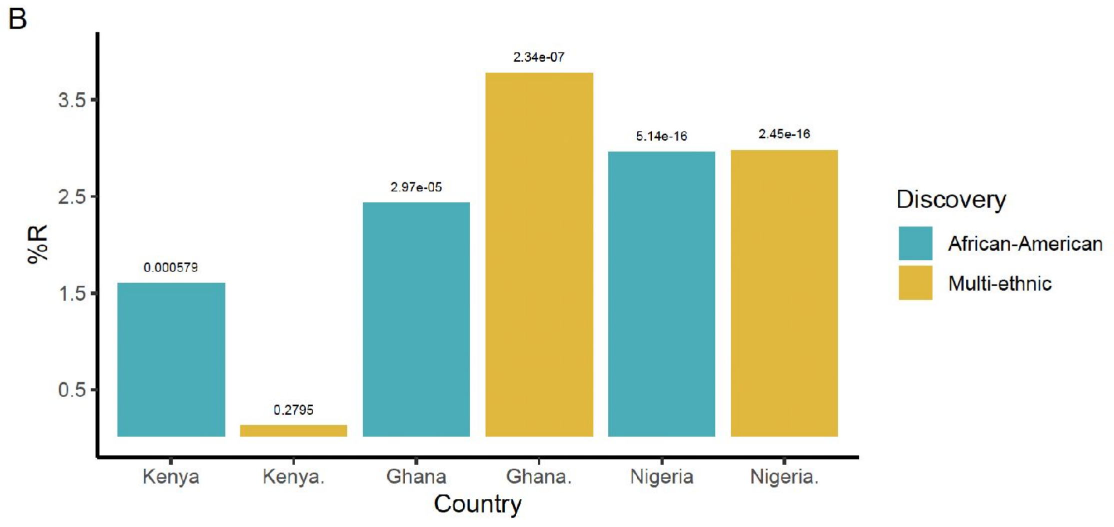
PRS Stratification and Transferability in African Countries

- The participants in the 10th decile of the African Ancestry–derived PRS had **more than threefold higher risk for developing T2D** per risk allele, compared with those in the first decile (OR 3.63; 95% CI 2.19–4.03; P = 2.79×10^{-17})
- On average, participants in the 10^{th} decile of the African Ancestry PRS were diagnosed with T2D 2.6 years earlier (*Beta* = 2.61; P=0.046) than participants in the first decile

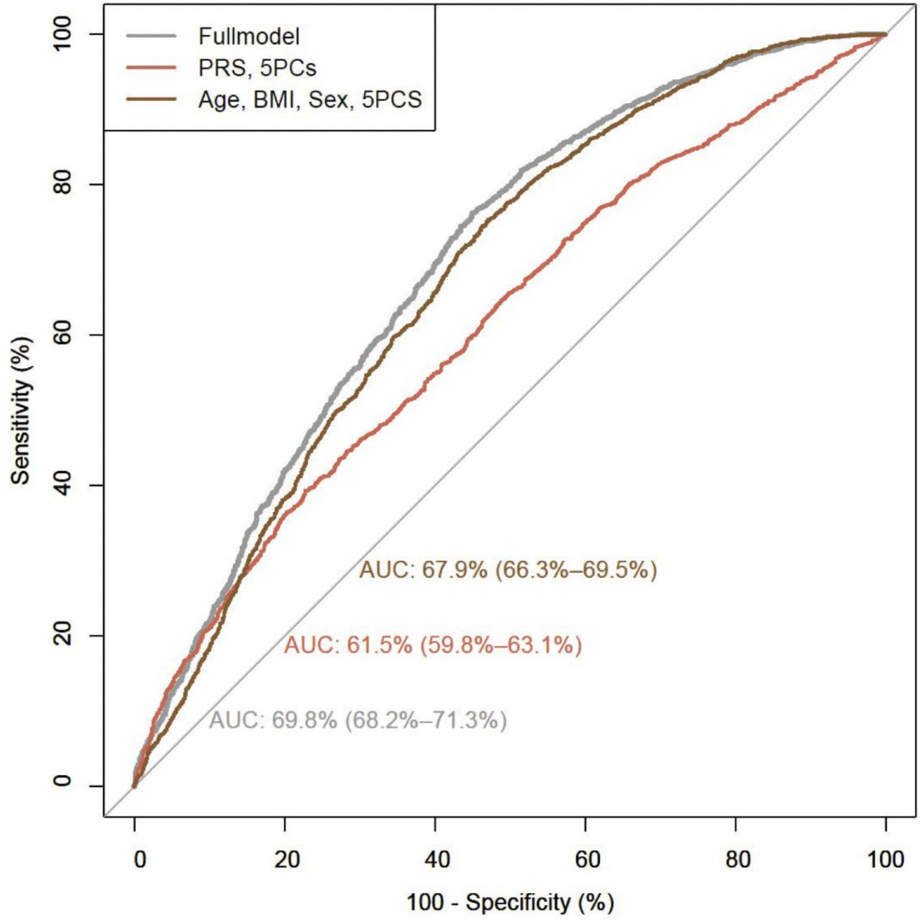
(

- The African Ancestry PRS was transferable in all countries compared with the multiethnic PRS that was not in Kenya.
- The PRS predictability (indicated by R2) varied greatly between the East Africa and West Africa countries, where predictability was much higher for both the African American and the multi-ancestry PRSs.





Discriminatory ability of the polygenic risk score



Chikowore T et al., … **Fatumo S** (2022) *Diabetes Care* 2022;45(3):717–723

- Model with the conventional risk factors of age, BMI, five PCs and sex had an area under the curve (AUC) of **67.9%**
- The African American PRS, five PCs, age, BMI and sex was **69.8%**
- This shows an **improved discriminatory ability by 1.9%,** with the addition of the African American PRS to the conventional risk factors.

Transferability of genetic risk scores of lipid traits in Africa

nature medicine

BRIEF COMMUNICATION

https://doi.org/10.1038/s41591-022-01835-x

Check for updates

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Transferability of genetic risk scores in African populations

Abram B. Kamiza[1,2,3], Sounkou M. Toure[1,4], Marijana Vujkovic[5], Tafadzwa Machipisa[6,7,8], Opeyemi S. Soremekun[1], Christopher Kintu[1], Manuel Corpas[9,10,11], Fraser Pirie[12], Elizabeth Young[13], Dipender Gill[14,15], Manjinder S. Sandhu[14], Pontiano Kaleebu[16], Moffat Nyirenda[16], Ayesha A. Motala[12], Tinashe Chikowore[3,17,20] ✉ and Segun Fatumo[1,16,18,19,20] ✉

Study design

- **Discovery (Base) Datasets**

  - *T2D **Multi-ancestry** meta-analysis GWAS (n= 1,407,282)*
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C+T Approach PRS using PRSice software

Target Dataset

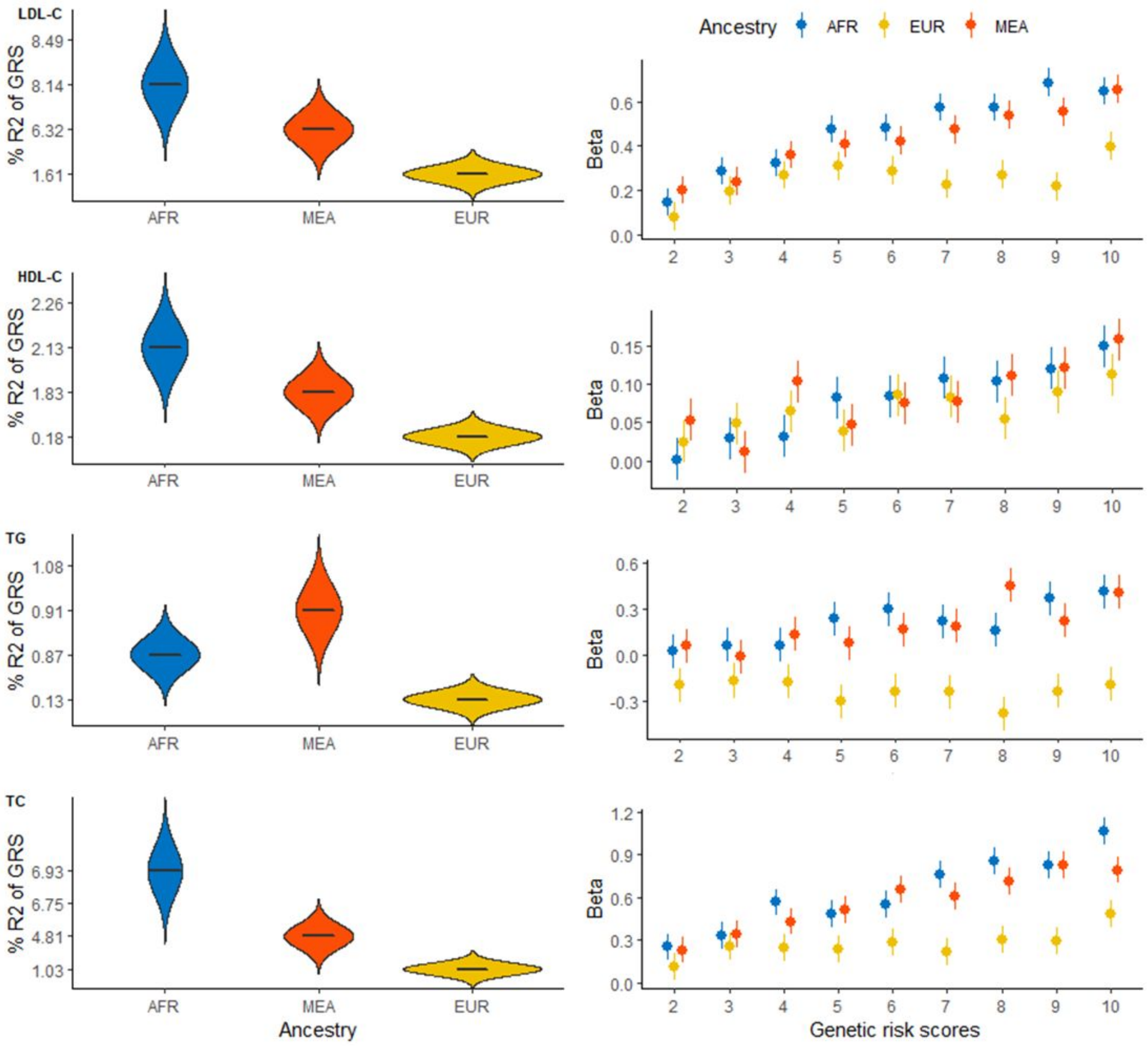
South Africa – Zulu
(*n=2,583*)

Validation Dataset

UGR
(*n=6407*)

We set out to evaluate best PRS to improve polygenic prediction in continental Africans.

Polygenic prediction of <u>lipid traits</u> in SSA (South Africa) compared to European and multi-ancestry scores.



- PRSs derived from African Ancestry individuals enhance polygenic prediction of <u>lipid traits</u> in SSA (Zulu south Africa cohort) compared to European and multi-ancestry scores.

The best-performing GRSs for LDL-C was African American (R^2 = 8.14%, P-value threshold (P_T) < 5×10^{-8}), followed by the multiancestry approach (derived from individuals of African ancestry, European ancestry and Hispanic American) (R^2 = 6.32%, $P_T < 5 \times 10^{-08}$), and the one from individuals of European ancestry (R^2 = 1.61%, $P_T < 5 \times 10^{-08}$

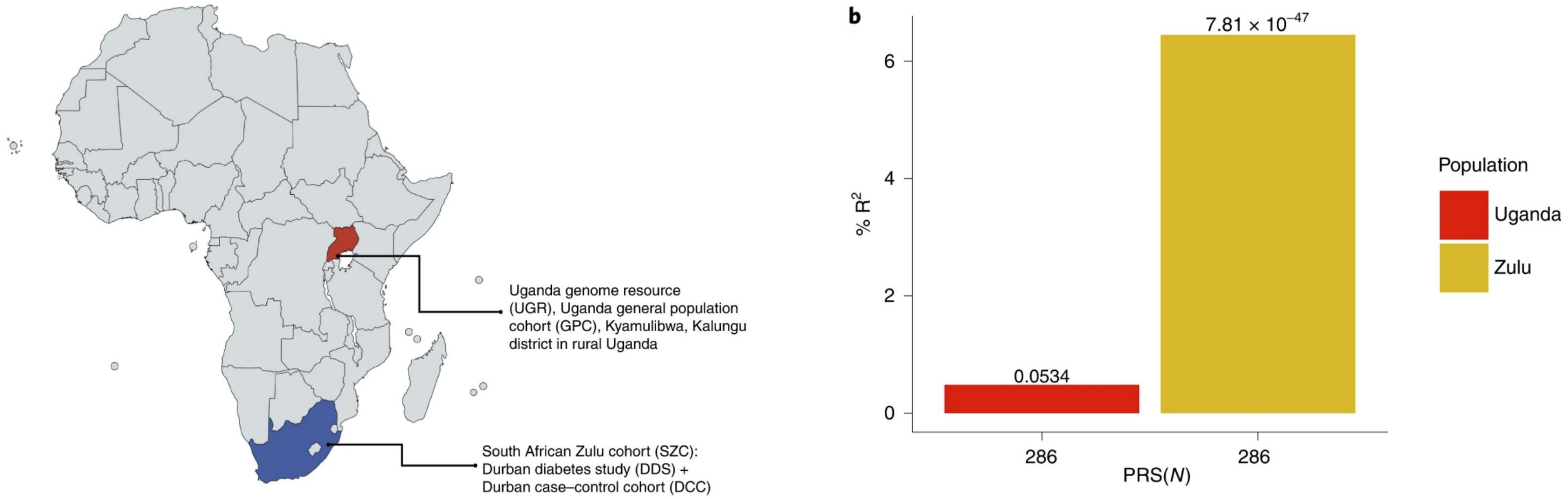
Poor predictive polygenic risk scores Performance across different lipid traits in Uganda Cohort

TableS2. Best predictive polygenic risk scores of lipid traits in Uganda replication cohort

Lipid traits	Populations	Adjusted R^2 (%)	Crude R^2 (%)	Predictive R^2	P-value†	exit	SE	SNPs	P-value
HDL-C	AFR	9.556	9.553	0.003	5×10^{-08}	-0.8021	1.731	173	0.6432
	EUR	9.562	9.553	0.009	5×10^{-08}	14.171	17.57	1,053	0.4201
	MEA	9.614	9.553	0.061	0.0046	7.0337	3.379	2,427	0.0374
LDL-C	AFR	12.663	12.637	0.026	5×10^{-08}	-6.742	4.909	271	0.1696
	EUR	12.692	12.637	0.055	5×10^{-08}	-48.158	24.18	774	0.0464
	MEA	12.662	12.637	0.025	5×10^{-08}	-27.288	20.43	1,008	0.1817
TG	AFR	2.737	2.735	0.002	5×10^{-05}	2.2983	7.591	720	0.7620
	EUR	2.786	2.735	0.051	0.0529	-97.963	53.41	122,574	0.0666
	MEA	2.782	2.735	0.047	0.0027	4.3846	2.494	491	0.0788
TC	AFR	17.439	17.391	0.048	5×10^{-08}	-20.679	10.71	286	0.0534
	EUR	17.393	17.391	0.002	5×10^{-08}	20.809	51.18	864	0.6843
	MEA	17.431	17.391	0.04	0.0268	9.1977	5.165	1,095	0.0751

† P-value threshold, SE, standard error; SNPs, single nucleotide polymorphisms; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; TG, triglycerides; TC, total cholesterol; AFR, African ancestry; EUR, European ancestry; MEA, multiethnic ancestry; MAA, multivariate of the African American.

- Comparative performance of polygenic prediction of Total Cholesterol using **the same GRS comprising 286 SNPs performed poorly in the Ugandan cohort (R^2 = 0.045%) but much better in the South African Zulu cohort (R^2 =6.345%)**



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Correspondence | Published: 23 February 2023

African genomes hold the key to accurate genetic risk prediction

Segun Fatumo & Michael Inouye

Nature Human Behaviour **7**, 295–296 (2023) | Cite this article

To achieve accurate genetic risk prediction and transferability across all populations, large-scale African data are necessary for PRS to achieve their potential.

Application, Challenges and Future Direction for the implementation of Polygenic Risk Score in Africa

Recommended Reading

Review | Open access | Published: 30 October 2023

Polygenic risk scores for disease risk prediction in Africa: current challenges and future directions

Segun Fatumo, Dassen Sathan, Chaimae Samtal, Itunuoluwa Isewon, Tsaone Tamuhla, Chisom Soremekun, James Jafali, Sumir Panji, Nicki Tiffin & Yasmina Jaufeerally Fakim

Genome Medicine **15**, Article number: 87 (2023) | Cite this article

https://genomemedicine.biomedcentral.com/articles/10.1186/s13073-023-01245-9

Full name (Study)	Country	Ancestry/ethics	Number of samples	Disease/traits	Lifestyle/exposure variables	Genotyping platform	Reference	Controlled access on EGA/DbGAP/ accession number
Uganda Genome Resource (UGR)	Uganda	Ugandans	7 000	Blood Pressure, lipid, Anthropometries, Full blood count, Liver Function, HbA1c, Kidney function, others	smoking, diet, physical activity, others	Illumina HumanOmni 2.5 M BeadChip array	[27, 28]	EGA: EGAD00010000965; EGAD00001001639
Durban Diabetes Study (DDS)	South Africa	South African Zulu	1,200	Diabetes Mellitus, BP, lipid, Anthropometries, Full Blood Counts, Liver Function	smoking, diet, physical activity, others	Customised Illumina Multi-Ethnic Genotyping Array (MEGA)	[29]	Not known
Durban Diabetes Case Control study (DCC)	South Africa	South African Zulu	1 600	Diabetes Mellitus, BP, lipid, Anthropometries, Full Blood Counts, Liver Function	smoking, diet, physical activity, others	Customised Illumina Multi-Ethnic Genotyping Array (MEGA)	[29]	Not known
Africa America Diabetes Mellitus (AADM)	Nigeria, Ghana & Kenya	Sub-Saharan African	5 000	Diabetes Mellitus, BP, lipid, Anthropometries, Full blood count, Liver Function, Kidney function, others	smoking, diet, physical activity, others	Affymetrix Axiom PANAFR SNP array	[30]	dbGAP: phs001844

Research on Obesity and Diabetes among African Migrants (RODAM and HELIUS)	Ghana, Suriname, Morocco	Africans	13 655	Diabetes Mellitus, BP, lipid, Anthropometries, Kidney Function, others	smoking, diet, physical activity, others	Not Known	[31]	EGA: EGAD00001004106
Muhimbili Sickle Cohort (Haemoglobin GWAS study)	Tanzania	Tanzania	1213	Sickle cell disease, Anthropometries, Others		Illumina Omni2.5 array	[32]	Not known
Genomic and Environmental Risk Factors for Cardiometabolic Disease in Africans (AWI-GEN)	Burkina Faso, Ghana, Kenya and South Africa	Sub-Saharan African	12 000	Blood Pressure, lipid, Anthropometries, Full blood count, Liver Function, Kidney function, others	smoking, diet, physical activity, others	H3Africa Illumina Array	[33, 34]	EGA EGAD00001004448
H3Africa Diabetes study (H3Africa Diabetics)	Various across Sub-Sahara Africa	Sub-Saharan African	12 621	Diabetes Mellitus, BP, lipid, Anthropometries, others	smoking, diet, physical activity, others	Planned H3Africa Illumina Array	[35]	Not Known
Genetics of rheumatic heart disease (RHDGen) Network	Various across Sub-Sahara Africa	Sub-Saharan African	3 555	rheumatic heart disease, BP, lipid, Anthropometries, others		Planned H3Africa Illumina Array	[36]	Not known
African Collaborative Center for Microbiome and Genomics Research (ACCME)	Nigeria	Nigerian	11 700	cervical cancer, Blood Pressure, lipid, Anthropometries, others	smoking, diet, physical activity, others	H3Africa Illumina Array	[37]	dbGAP phs001945
African Female Breast Cancer Epidemiology (AFBRECANE)	Nigeria	Nigerian	1 000	Breast Cancer, Blood Pressure, lipid, Anthropometries, others	smoking, diet, physical activity, others	Not Known	[38]	Not known
KEMRI VaccGene study	Kenya	Kenya	1 400	Blood Pressure, Anthropometries, others	smoking, diet, physical activity,	Not Known	[39, 40]	Not known

Malawi Genome Resource	Malawian	Malawian	6 626	Diabetes Mellitus, BP, lipid, Anthropometries, Kidney Function, others	smoking, diet, physical activity, others	H3Africa Illumina Array version 2	[41]	Not known
The Stroke Investigative Research and Educational Network (SIREN)	Nigeria	Nigerian	8 000	Stroke, Blood Pressure, Anthropometries and others	smoking, diet, physical activity, others	H3Africa Illumina Array	[42]	Not known
Trauma and PTSD study	Rwanda	Rwandans	400	Trauma, Post-traumatic stress disorder (PTSD), Blood Pressure, Anthropometries and others	smoking, diet, physical activity, others	Planned H3Africa Illumina Array	[43, 44]	Not known
Keneba Biobank	Gambia	Gambians	10 000	Blood Pressure, serum creatinine, Anthropometries and others	smoking, diet, physical activity, others	Planned H3Africa Illumina Array	[45]	Not known
The Sickle Cell Disease Genomics of Africa (SickleGenAfrica)	Cameroon, Ghana, Nigeria and Tanzania	Africans	7 000	Sickle cell disease, Blood Pressure, Anthropometries and others	smoking, diet, physical activity, others	H3Africa Illumina Array	[46]	Not known
The H3Africa Kidney Disease study	Ghana, Nigeria, Ethiopia, and Kenya	Africans	11 964	Kidney function, Blood Pressure, Anthropometries and others	smoking, diet, physical activity, others	Planned H3Africa Illumina Array	[47]	EGA EGAD00010002365; EGAD00001009333

Collaborative African Genomics Network (CAfGEN)	Botswana, Uganda, and Swaziland	Africans	1 000	HIV, Blood Pressure, Anthropometries and others	smoking, diet, physical activity, others	Illumina HiSeq 2500	[48]	EGAD00001006224
The Tanzania Stroke Incidence Project (TSIP)	Tanzania	Tanzanian	636	Stroke, Anthropometries, Others		Not Known	[49]	Not known
Non-Communicable Diseases – Genetic Heritage Study (NCD-GHS)	Nigeria	Nigerian	100 000	Blood Pressure, lipid, Anthropometries, Full blood count, Liver Function, HbA1c, Kidney function, Cancer, Diabetes Mellitus others	smoking, diet, physical activity, others	Illumina Global Array	[50]	Not known
H3AFRICA TRYPANOGEN PROJECT	Uganda Guinea Ivory Coast Burkina Faso Cameroon DRC Malawi Zambia	Africans	233	trypanosomiasis	lifestyle	Illumina HiSeq 2500	[51]	EGAS00001002602

H3Africa Genomic Characterization and Surveillance of Microbial Threats in West Africa	Nigeria, Sierra Leone	West Africa	2667 1345	Lassa Fever	lifestyle	Illumina Omni 2.5 M, Illumina Omni 5 M	[52]	EGAD00010002509
NeuroGAP-Psychosis	Kenya, Ethiopia, Uganda, South Africa	South and East Africans	17,000	Schizophrenia, Bipolar disorder	lifestyle	Global Sequencing Array	[53]	dbGaP accession phs002528.v1.p
NeuroDev project	Kenya, South Africa	South and East Africans	5,000	Autism spectrum disorder (ASD), Intellectual disability (ID), Attention deficit hyperactivity disorder (ADHD)	lifestyle	Global Sequencing Array	[54]	dbGap accession number phs001874
The African Biobank	Nigeria, Kenya, Tanzania, Zimbabwe and South Africa	Africans	1,488	Pharmacogenetics	lifestyle	PCR–RFLP	[55]	Not known
MalariaGEN	Burkina Faso, Cameroon, Ghana, Kenya, Malawi, Mali, Nigeria, Papua New Guinea, Tanzania, The Gambia, Vietnam	Africa, Asia and Oceania	17,000	Malaria	lifestyle	Sequenom MassArray platform	[56–58]	EGAS00000000026

There are many factors contributing to poor transferability of PRS in African populations. This includes genetic factors such as	minor allele frequencies,
	difference in linkage disequilibrium patterns, and
	their interactions with environmental considerations like diet, exercise, age, gender, and variability in phenotype measurement.

Applications of Polygenic Risk Scores

Disease Prediction

PRS can be used to identify individuals at high risk for certain diseases, enabling early intervention and preventive strategies.

Personalized Screening

PRS can guide personalized screening programs, such as targeted cancer screenings, based on an individual's genetic risk profile.

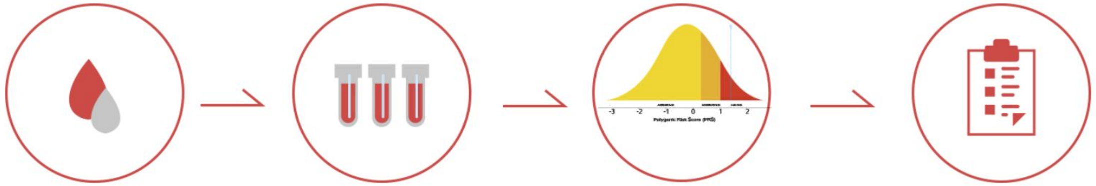
Drug Response

PRS can help predict an individual's response to certain medications, allowing for more targeted and effective treatments.

KARDIOGEN: Preventive Healthcare using PRS test for Coronary Artery Disease (CAD) in india

What is the test process?

Just 3 ml of your blood can give your CAD PRS score



3 ml blood → Lab run → PRS Prediction → Report with CAD Risk Score (High / Moderate / Average)

Results within 12 working days

What does one do after receiving your CAD score

CAD Risk Profile	What it means?	Action
High	You are at a higher genetic risk (3 times)of getting CAD than a average risk counterpart.	Take a comprehensive heart check up and consult your doctor for an invasive procedure
Moderate	You are at a higher genetic risk (~1.5 times) towards CAD than a average risk counterpart	Avoid all lifestyle risks such as smoking, stress, etc.Maintain normal blood pressure and blood sugar levels with regular monitoring. Medication, such as blood thinners can be considered.
Average	You are at a average genetic risk towards coronary events	Maintain a healthy lifestyle and get routine health checkups.

Example: Polygenic Risk Score for Coronary Artery Disease

Genetic Markers

Hundreds of genetic variants have been associated with an increased risk of coronary artery disease.

Risk Stratification

PRS can stratify individuals into high, intermediate, and low-risk categories for coronary artery disease.

Preventive Measures

Individuals with high PRS for coronary artery disease can be targeted for more intensive screening and lifestyle interventions.

Clinical Utility

PRS for coronary artery disease has shown promise in improving risk prediction and guiding preventive strategies.

Limitations and Challenges of Precision Medicine in Africa

1. Infrastructure and Resources:

- **Limited Healthcare Infrastructure**: Many African countries lack the necessary healthcare infrastructure, including **advanced diagnostic labs and reliable healthcare facilities, EHR** to implement precision medicine.
- **Funding Constraints**: The high cost of genomic sequencing and precision treatments is a significant barrier, given the limited healthcare budgets in many African nations.

2. Data and Technology:

- **Insufficient Data**: There is a lack of comprehensive genomic data from African populations, which is crucial for developing effective precision medicine strategies.
- **Technological Barriers**: Limited access to advanced technologies and bioinformatics tools hinders the collection, analysis, and interpretation of genetic data.

Limitations and Challenges of Precision Medicine in Africa

3. Regulatory and Ethical Issues:

- **Regulatory Frameworks**: Many African countries lack robust regulatory frameworks to govern genetic research and the application of precision medicine.

- **Ethical Concerns**: Issues such as genetic privacy, informed consent, and potential misuse of genetic information need to be carefully addressed to protect individuals and communities.

4. Socioeconomic and Cultural Factors:

- **Health Literacy**: Low levels of health literacy and awareness about precision medicine (even among the most educated health practitioner) can impede its acceptance and adoption among the population.

- **Cultural Beliefs**: Traditional beliefs and skepticism about modern medical practices can influence the acceptance of precision medicine approaches.

Limitations and Challenges of Precision Medicine in Africa

5. Equity and Access:

- **Healthcare Disparities**: Socioeconomic disparities can limit access to precision medicine, with rural and underserved communities being particularly disadvantaged.

- **Equitable Distribution**: Ensuring equitable access to the benefits of precision medicine across different regions and populations in Africa is a significant challenge.

Future Potential of Precision Medicine in Africa

1. Advancements in Genomic Research:

- **Diverse Genetic Data**: Africa's genetic diversity offers a unique opportunity to advance genomic research, leading to discoveries that could benefit global health.

- **Research Initiatives**: Increasing genomic research initiatives, such as the African Genome Project, can provide valuable insights into the genetic basis of diseases.

Future Potential of Precision Medicine in Africa

2. Improved Disease Management:

- **Personalized Treatment Plans**: Tailoring treatment plans based on genetic information can improve the efficacy and reduce side effects of treatments for conditions like cancer, cardiovascular diseases, and diabetes.

- **Targeted Therapies**: Precision medicine can enable the development of targeted therapies for prevalent diseases in Africa, such as malaria, tuberculosis, and HIV/AIDS.

Future Potential of Precision Medicine in Africa

3. Public Health Interventions:

- **Preventive Healthcare**: Genomic screening can help in early detection and prevention of hereditary diseases, reducing the overall healthcare burden.

Population Health Management: Precision medicine can aid in the development of more effective public health interventions by identifying genetic and environmental risk factors for diseases prevalent in specific communities.

Future Potential of Precision Medicine in Africa

4. Capacity Building and Education:

- **Training and Development**: Investment in training healthcare professionals and researchers in genomics and precision medicine can build local capacity and expertise.
- **Collaborative Research**: International collaborations and partnerships can facilitate knowledge transfer and resource sharing.

Conclusion

- PRS holds real promise for precision medicine in Africa, but current performance is fundamentally limited by the lack of large, well-powered African GWAS datasets.

- Poor transferability of European and trans-ancestry PRS in Africa is not a failure of the method, but a reflection of Africa's genetic diversity and distinct disease architecture.

- PRS performance varies substantially across African regions, highlighting the need for geographically diverse discovery cohorts rather than a single "African" reference.

- African ancestry–derived PRS consistently outperform European PRS for kidney function, diabetes, and lipid traits, reinforcing the urgency of building African genomic resources.

- Scaling African genomic studies is no longer optional but essential for equitable precision medicine, global risk prediction, and improved clinical translation.

- Partnership, training, capacity building, and data sharing through initiatives like KidneyGenAfrica are the fastest route to closing the precision medicine gap and ensuring Africa benefits from PRS innovation.

Acknowledgements

Group members in the UK and Uganda

https://www.genomicdiversity.org/

wellcome

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KidneyGen Africa

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H3Africa Human Heredity & Health in Africa

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Thank You